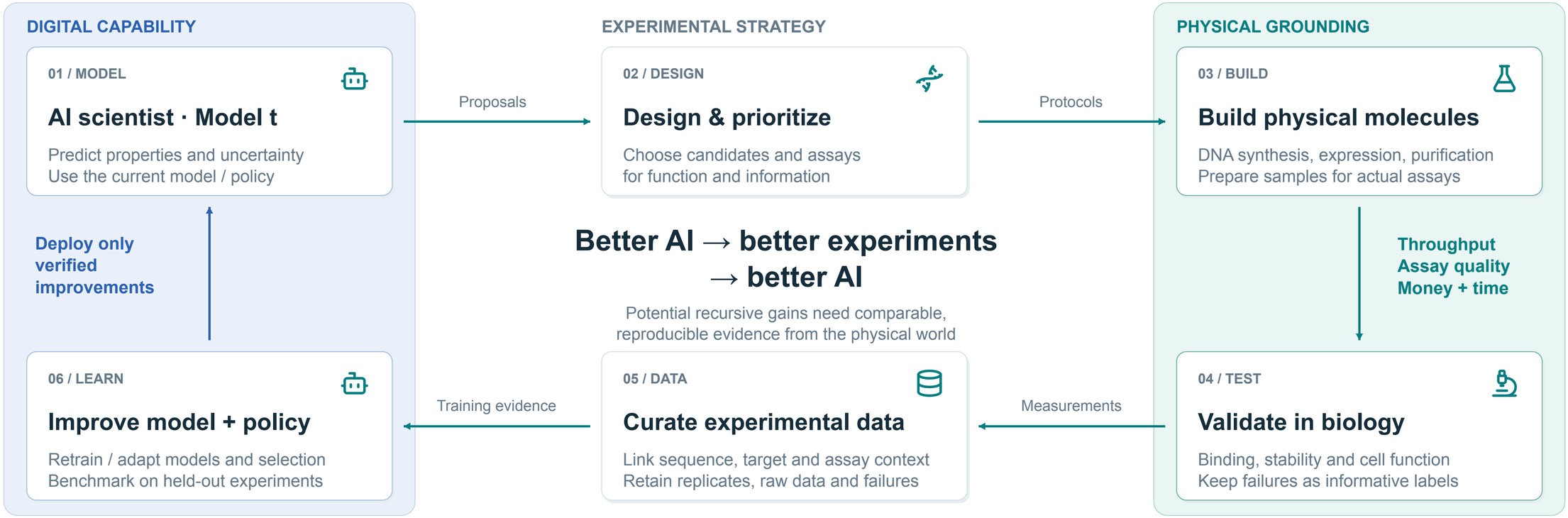
RSI: AI design and biological validation

How experimental feedback could improve both biological designs and the AI design process

RECURSIVE SELF-IMPROVEMENT · CONCEPTUAL MECHANISM



LST supports every return to physical biology

BUILD
DNA synthesis · expression · purification

TEST
Proteins · biosensors · analytical columns
Cell assays · measurement platforms

CONNECT + SCALE
Labware · automation · data platforms
Comparable records across batches

The pace of improvement depends on useful experimental evidence per cycle

Potential LST demand ∝ funded cycles × assay depth × net inputs per assay

Automation and miniaturization can reduce inputs per measurement; instrument CAPEX is separate.

Conceptual RSI mechanism. Closed-loop learning does not establish sustained autonomous RSI. Primary sources: SOURCES.md.